

Mechanically Versatile Soft Machines through Laminar Jamming

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There are two major structural paradigms in robotics: soft machines, which are conformable, durable, and safe; and traditional rigid robots, which are fast, precise, and capable of applying high forces. Here, the paradigms are bridged by enabling soft machines to behave like traditional rigid robots on command. This task is accomplished via laminar jamming, a structural phenomenon in which a laminate of compliant strips becomes strongly coupled through friction when a pressure gradient is applied, causing dramatic changes in mechanical properties. Rigorous analytical and finite element models of laminar jamming are developed, and jamming structures are experimentally characterized to show that the models are highly accurate. Then jamming structures are integrated into soft machines to enable them to selectively exhibit the stiffness, damping, and kinematics of traditional rigid robots. The models allow jamming structures to efficiently meet arbitrary performance specifications, and the physical demonstrations illustrate how to construct systems that can behave like either soft machines or traditional rigid robots at will, such as continuum manipulators that can rapidly have joints appear and disappear. This study aims to foster a new generation of mechanically versatile machines and structures that cannot simply be classified as “soft” or “rigid.”

1. Introduction

Soft machines and traditional rigid robots have distinct forms and functions. Soft machines (e.g., elastomeric bending actuators^[1,2] and dielectric elastomer grippers^[3,4]) are made of compliant materials and bend or twist continuously along their length. Their actuation mechanism is typically distributed throughout their volume. Traditional rigid robots (e.g., robotic arms and humanoids) are made of stiff materials and bend or translate discretely at joints. Their actuation mechanism is usually confined to these joints. The structure of soft machines allows them to conform to complex shapes,^[5,6] withstand crushing loads,^[7] dampen impacts, and interact safely with the body.^[8,9] In contrast, the structure of traditional rigid robots enables them to perform tasks quickly, precisely, and with high

resolution, as well as resist deformation, apply high forces, and oscillate with minimal decay.

To make more versatile robots, researchers have aimed to enable soft machines to selectively behave like traditional rigid robots. In particular, soft machines have been constructed with materials and structures that can exhibit tunable stiffness and damping in order to attain the mechanical properties of traditional robots. These components include low-melting-point materials,^[10,11] shape-memory materials,^[12,13] magnetorheological (MR) fluids,^[14] and granular structures.^[9,15] Nevertheless, most of these technologies cannot achieve a wide range of stiffness and damping values per unit weight (MR fluids, granular structures), have low resolution of stiffness and damping values (low-melting-point materials), transition between these values slowly (low-melting-point materials, shape-memory materials), and/or have poor resistance to bending moments

(MR fluids, granular structures).^[9,16] Furthermore, none of these technologies have yet enabled devices to rapidly transition between the continuous deformations typical of soft machines and the discrete, jointed deformations typical of traditional robots. Previous efforts toward this goal have been limited by slow transition times, high stiffnesses when no power is applied, and low efficiencies.^[10,17]

The laminar jamming (a.k.a., “layer jamming”) phenomenon is a promising alternative to these technologies (Figure 1A,B). Laminar jamming structures are lightweight and can be rapidly actuated; moreover, they can achieve excellent range and resolution of stiffness and damping values with high resistance to bending moments. A laminar jamming structure consists of a laminate of flexible strips or sheets. In its default state, the laminate is highly compliant. However, when a pressure gradient is applied (in this study, by enclosing the laminate in an airtight envelope and applying a vacuum to the envelope), increased frictional interactions dramatically augment the bending stiffness of the structure; in addition, at high loads, the structure dissipates energy. Researchers have applied laminar jamming to haptics,^[18–20] medical devices,^[21,22] and soft actuators.^[23,24] Nevertheless, these studies have not yet provided analytical or computational models for laminar jamming beyond an initial deformation phase, making design of practical jamming structures an arduous process. Furthermore, they have not

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 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/adfm.201707136>.

DOI: 10.1002/adfm.201707136

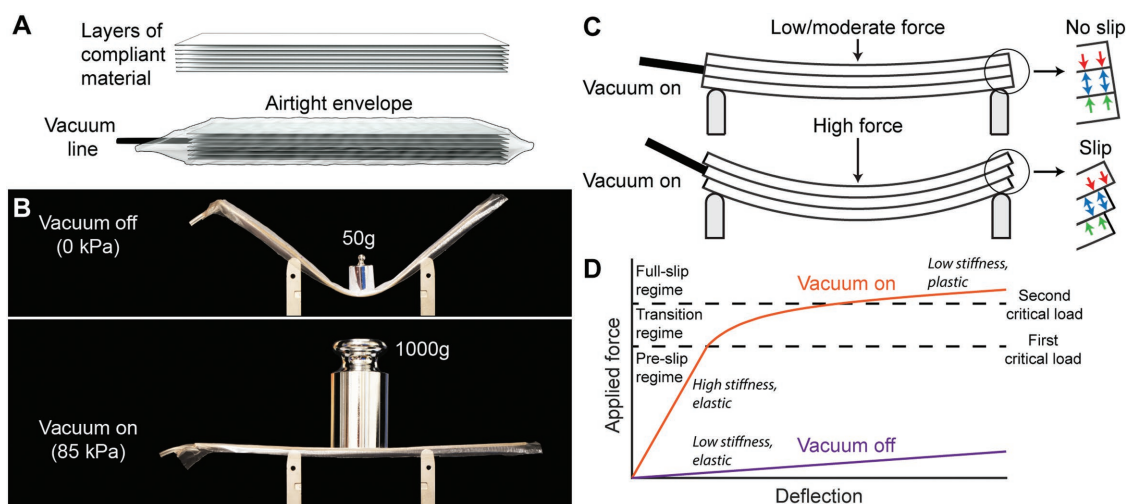


Figure 1. Fundamental behavior of laminar jamming structures. A) Schematic of a jamming structure. B) When vacuum is off, the layers bend independently, and the structure has low bending stiffness. When vacuum is on, the layers bend as a cohesive unit, and the structure has high bending stiffness. C) However, when vacuum is on, the layers are cohesive only until a critical force. For higher forces, longitudinal shear stress is large enough to cause the layers to slip at certain points along their interfaces. D) Summary of mechanical behavior. When vacuum is off, the structure has low bending stiffness, which is proportional to the slope of the curves. When vacuum is on, the structure has three deformation regimes. In pre-slip, the bending stiffness is maximal and constant. After the first critical load, the structure enters the transition regime, in which the layers begin to slip. The bending stiffness decreases. After the second critical load, the structure enters full-slip, in which the layers have slipped at all possible points along their interfaces. The bending stiffness is minimal and constant. When slip occurs, energy is dissipated to friction between the layers, and the structure behaves plastically.

yet explored how laminar jamming can be used to transform bending kinematics.

In this paper, we model laminar jamming in detail and demonstrate how the technology can bridge the gap between soft machines and traditional rigid robots. Specifically, we develop an analytical model that mathematically captures how two-layer jamming structures behave over all major phases of deformation. We then develop finite element models that extend these predictions to many-layer jamming structures, as well as describe how their stiffness and damping depend on critical design inputs (e.g., the vacuum pressure). These models are validated through rigorous experimental characterization. Together, the analytical and finite element models present researchers with the first means to rapidly and accurately design jamming structures to meet arbitrary design requirements.

We then demonstrate the capabilities of laminar jamming structures by integrating them into real-world pneumatic and cable-driven soft machines. In the process, we achieve two novel functions that illustrate how these machines can reversibly emulate traditional rigid robots: (1) shape-locking, in which a compliant system can selectively manifest a stiff version of a desired shape and preserve it, even after powering off the actuators, and (2) variable kinematics, in which a compliant system can transition between continuous bending and discrete, jointed bending on command. The variable kinematics function is then used to build a two-fingered grasper that can perform pinch grasps on small objects, as well as wrap grasps on objects of eight times the diameter. These demonstrations prove the feasibility of using laminar jamming to build mechanically versatile machines and structures that exhibit both soft and traditional behaviors.

2. Results

2.1. Analytical Modeling

As described earlier, when a vacuum is applied to a laminar jamming structure, the bending stiffness increases dramatically. Previous studies have shown that the stiffness increases by a factor of n^2 , where n is the number of layers in the structure; thus, applying a vacuum to a structure with just 32 layers can increase its stiffness by 3 orders of magnitude. However, the vacuumed jamming structure sustains this increased stiffness only over a limited range of loads, beyond which the stiffness declines.^[18,21]

In our investigation, physical reasoning suggested that this behavior reflected three phases of deformation in a vacuumed jamming structure (Figure 1C,D): (1) In pre-slip, the layers are cohesive, and the stiffness of the structure is a factor of n^2 greater than the stiffness without vacuum. No energy is dissipated, and the damping (i.e., dissipated energy per unit deflection) is zero. As the structure is loaded, the longitudinal shear stress along the interfaces between layers begins to rise. (2) In the transition regime, the longitudinal shear stress along certain regions of the interfaces equals the maximum possible shear stress, which is determined by the coefficient of friction and the pressure gradient. Layers begin to slip along those regions, and the stiffness of the structure decreases. Energy is dissipated to friction, and the damping increases. (3) In full-slip, all layers have slipped along the full length of their interfaces. The stiffness of the structure is minimal, and the damping is maximal.

To mathematically capture this behavior, we derived an analytical model that rigorously described the deformation and mechanical properties of jamming structures during these phases. Our model was based on Euler–Bernoulli beam theory;

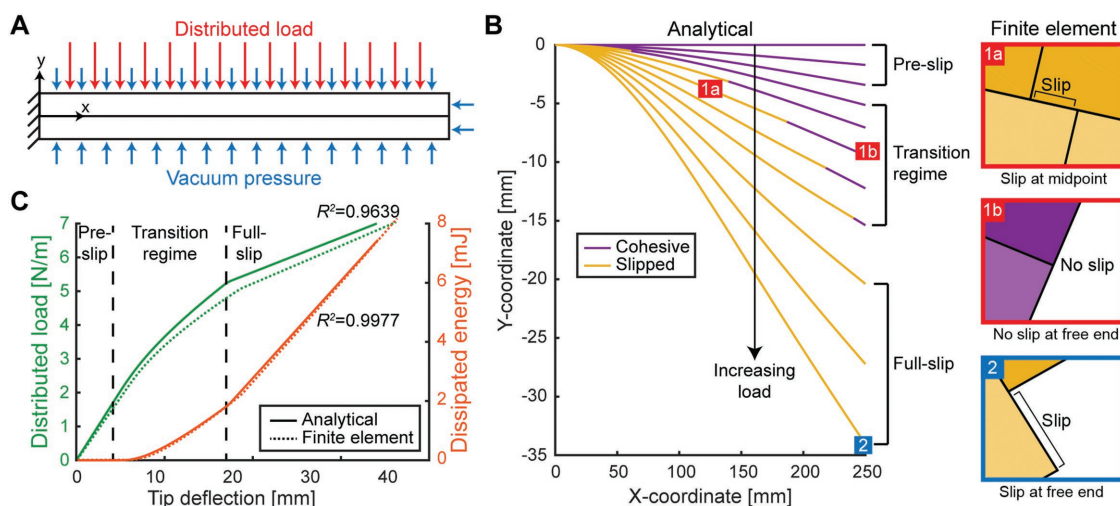


Figure 2. Analytical model of two-layer jamming structures. A) Schematic of example jamming structure. B) Elastica of jamming structure for increasing loads. The slipped region is highlighted at each load; because shear stress decreases along the x -direction, the slipped region initiates at the clamped end and grows toward the free end. Two-layer finite element models corroborated that slip occurred in analytically predicted regions. C) Bending stiffness is proportional to the slope of the load-versus-deflection curve; as expected, the stiffness transitions from a minimal to a maximal value. Damping is proportional to the slope of the dissipated-energy-versus-deflection curve; damping transitions from zero to a maximal value. Finite element models closely corroborated analytically predicted stiffness and damping values. (SI: Analytical Modeling: Case Study)

however, we extended the theory to describe how mechanical behavior was affected by vacuum pressure, friction at the interfaces between layers, and slip along the interfaces. Governing equations were derived using equilibrium and moment-stress relations, and general boundary conditions were formulated (SI: Analytical Modeling: Governing Equations and Boundary Conditions). The boundary-value problem was then solved for a two-layer cantilevered jamming structure under a uniform distributed load (SI: Analytical Modeling: Explicit Solution); this case was chosen to illustrate slip propagation (i.e., growth of the regions along which layers slip), which is exhibited by most jamming structures.

The model predicted the elastica (i.e., the shape), stiffness, dissipated energy, and damping of the jamming structure. The model also predicted the transition loads (i.e., the loads at which the jamming structure shifts from one deformation phase to the next), as well as the length of the region along which the layers slipped. Furthermore, it provided the functional dependence of all the preceding quantities on dimensions, material properties, the vacuum pressure, and the applied load (SI: Analytical Modeling: Summary of Formulae). For example, the model showed that the full-slip damping force was given succinctly by μPbh , where μ is the coefficient of friction, P is the vacuum pressure, b is the width of a layer, and h is the height. Dimensionless forms of the equations in the model were derived as well (SI: Analytical Modeling: Dimensionless Forms). The model was evaluated for an example structure (Figure 2), and the results were corroborated by two-layer finite element models (SI: Finite Element Modeling: Two-Layer Jamming Structures).

2.2. Finite Element Modeling and Experimental Characterization

Although the analytical model rigorously predicted the mechanical behavior of two-layer jamming structures, designers may

desire to build real-world jamming structures with additional layers to further adjust their properties. Our analytical model can be directly extended to describe many-layer jamming structures (SI: Analytical Modeling: Extending the Model). However, the process is algebraically taxing, and numerical methods may be preferred.

To predict the mechanical behavior of many-layer jamming structures, we conducted finite element simulations. The jamming structures were modeled as 2D plane-strain structures with dimensions, material properties, boundary conditions, and loads equal to those of real-world jamming structures used later in experimental validation (SI: Finite Element Modeling: Stiffness and Damping of Many-Layer Jamming Structures). Furthermore, simultaneous frictional contact was allowed to occur at all interfaces, and large-deformation analysis was enabled. No fitting parameters were used.

The results of the finite element simulations were used to quantify how critical design inputs affected major performance metrics of many-layer jamming structures. Specifically, the number of layers, vacuum pressure, and coefficient of friction of the layers were varied, and the stiffness and damping values of the jamming structures during pre-slip and full-slip were extracted. The polynomial relationship between each input and output was determined, and the resulting scaling relations were tabulated (SI: Finite Element Modeling: Functional Dependencies). For example, full-slip damping was found to scale linearly with number of layers, vacuum pressure, and coefficient of friction.

To evaluate the accuracy of the finite element models, experimental characterization of many-layer jamming structures was conducted. Jamming structures were fabricated according to a multistep process (SI: Experimental Characterization: Fabrication Process), and the repeatability of the structures was assessed (SI: Experimental Characterization: Repeatability Analysis). The jamming structures were highly repeatable

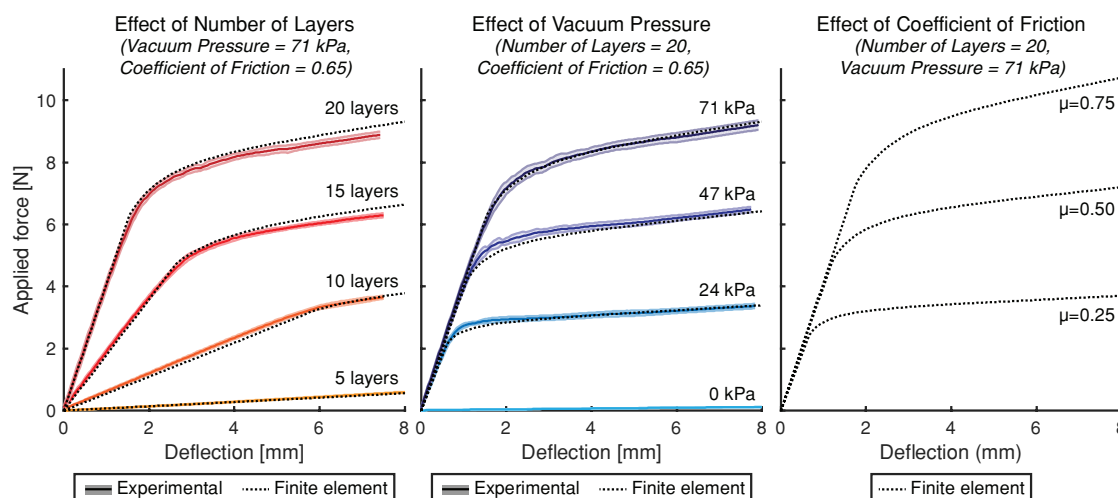


Figure 3. Finite element predictions and experimental validation for many-layer jamming structures. Jamming structures were loaded in three-point bending. Each experimental curve in fact consists of a mean curve and shaded error bar that spans ± 1 standard deviation; the maximum deviation on any curve is 0.24 N, indicating high repeatability. The minimum coefficient of determination (R^2) between finite element and experimental data is 0.9879, demonstrating exceptional predictive accuracy. (No experimental data are shown for coefficient of friction, as friction could not be precisely adjusted.) Hysteresis and damping predictions were experimentally evaluated as well (Figure S6, Supporting Information).

from loading cycle to loading cycle and sample to sample. The many-layer jamming structures were then tested in three-point bending for various numbers of layers and vacuum pressures (SI: Experimental Characterization: Stiffness and Damping Characterization Process). Transverse force and maximum deflection was recorded, and finite element predictions were compared to experimental data (Figure 3). The finite element models predicted experimental results with exceptional accuracy.

2.3. Useful Functions

2.3.1. Shape-Locking

Two real-world capabilities of laminar jamming structures were demonstrated by integrating them into soft machines. First, the shape-locking function was demonstrated. A pneumatically powered soft bending actuator was fabricated (SI: Functions and Applications: Shape-Locking), and a 20-layer jamming structure was adhered to the ventral surface (i.e., the longitudinal surface closer to the center of curvature when the actuator was inflated). The actuator was then pressurized. When the actuator was depressurized, the system naturally returned to its undeformed configuration; however, when a vacuum was applied to the jamming structure before the actuator was depressurized, the system preserved its shape with high fidelity (Figure 4).

2.3.2. Variable Kinematics

Next, the variable kinematics function was demonstrated. A robotic system was designed that consisted of three major parts: a silicone rubber substrate, a three-part jamming structure (i.e., three stacks of material, separated by narrow gaps),

and a cable routed through the substrate to actuate bending (Figure 5A). Note that when the rubber substrate and the vacuumed state of the jamming structure are considered separately, their bending kinematics are entirely distinct. The substrate bends continuously along its length, whereas the vacuumed jamming structure bends discretely at its narrow gaps, which act as joints. When the substrate and the jamming structure are adhered, the bending kinematics of the system may vary between these two extremes.

To enable the system to transition between continuous and discrete kinematics, the bending stiffnesses of the substrate and jamming structure were judiciously selected. The thickness of the substrate was chosen so that $k_{\text{sub}} = (k_{\text{jam}}^{\text{nv}} * k_{\text{jam}}^{\text{v}})^{\frac{1}{2}}$, where k_{sub} is the bending stiffness of the substrate, $k_{\text{jam}}^{\text{nv}}$ is the stiffness of the jamming structure without vacuum, and $k_{\text{jam}}^{\text{v}}$ is the pre-slip stiffness of the jamming structure with vacuum. (In equivalent terms, k_{sub} was the geometric mean of the unjammed and jammed stiffnesses.) In addition, the number of layers in the jamming structure was chosen so that $k_{\text{jam}}^{\text{nv}} \gg k_{\text{jam}}^{\text{v}}$. Thus, when no vacuum was applied and the cable was pulled, the stiffness of the system would be dominated by k_{sub} , and the system would bend continuously. When vacuum was applied, the stiffness would be dominated by $k_{\text{jam}}^{\text{v}}$, and the system would bend discretely.

To evaluate this concept prior to prototyping, finite element simulations of the system were conducted (SI: Finite Element Modeling: Variable Kinematics). The system was modeled as a multipart 2D plain-strain structure fixed at one end, and to approximate cable loading, a pure moment load was applied at the free end. The shape of the system was visualized, and the ratio of maximum to mean curvature $\left(\frac{\kappa_{\text{max}}}{\kappa_{\text{mean}}} \right)$ was computed along the ventral arc as a measure of discreteness. When no vacuum was applied, the system deformed continuously, and

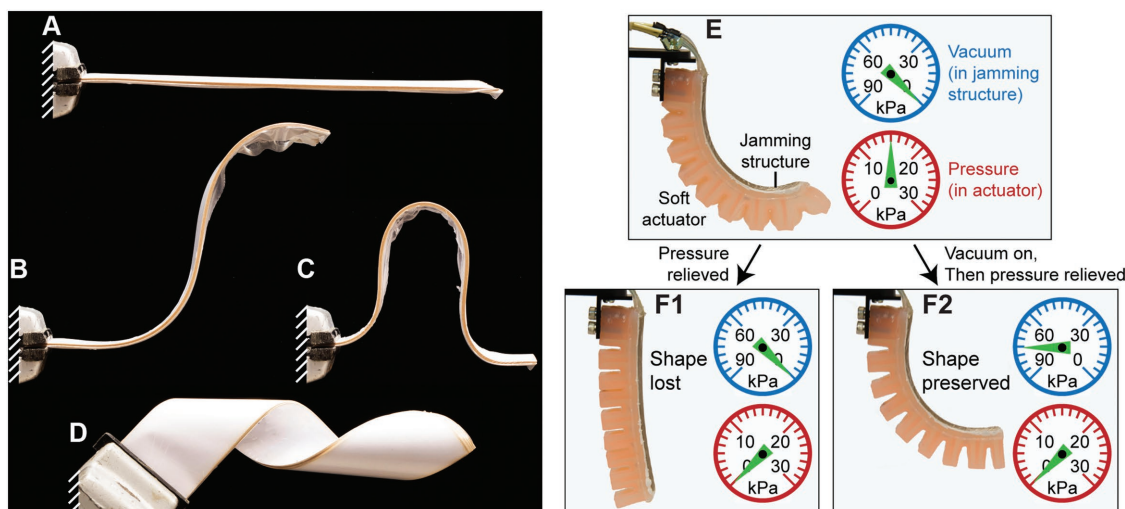


Figure 4. Demonstration of shape-locking function. A–D) A jamming structure consisting of 20 layers of copy paper was deformed into various shapes; vacuum was then applied, and the shape was preserved in all cases. E) A 20-layer jamming structure was then adhered to a pneumatic soft actuator. The actuator was initially pressurized to 16 kPa to achieve a desired bending angle. F1) In a first test, no vacuum was applied to the jamming structure, and the actuator was depressurized. The composite structure immediately returned to its undeformed state. F2) In a second test, a vacuum pressure of 85 kPa was first applied to the jamming structure, and the actuator was then depressurized. Quantity R^2 between the arc of the ventral surface in E and the same arc in F2 was 0.9835. Thus, the system preserved its shape with high fidelity.

$\frac{\kappa_{\max}}{\kappa_{\text{mean}}}$ remained low. When vacuum was applied, the system deformed discretely, and $\frac{\kappa_{\max}}{\kappa_{\text{mean}}}$ increased by a factor of 6.65 at high loads (Figure 5B,C).

Finally, a prototype of the system was fabricated (SI: Functions and Applications: Variable Kinematics). The prototype deformed according to finite element predictions, and application of vacuum allowed it to select between continuous and discrete kinematics (Figure 5D).

2.4. Application

2.4.1. Two-Fingered Grasper

In robotic hands, compliant fingers that bend continuously can facilitate wrap grasps around large objects,^[25] whereas rigid fingers that bend discretely at joints can facilitate pinch grasps around smaller objects;^[26,27] it is challenging to design and fabricate hands capable of both. To accomplish the task, we

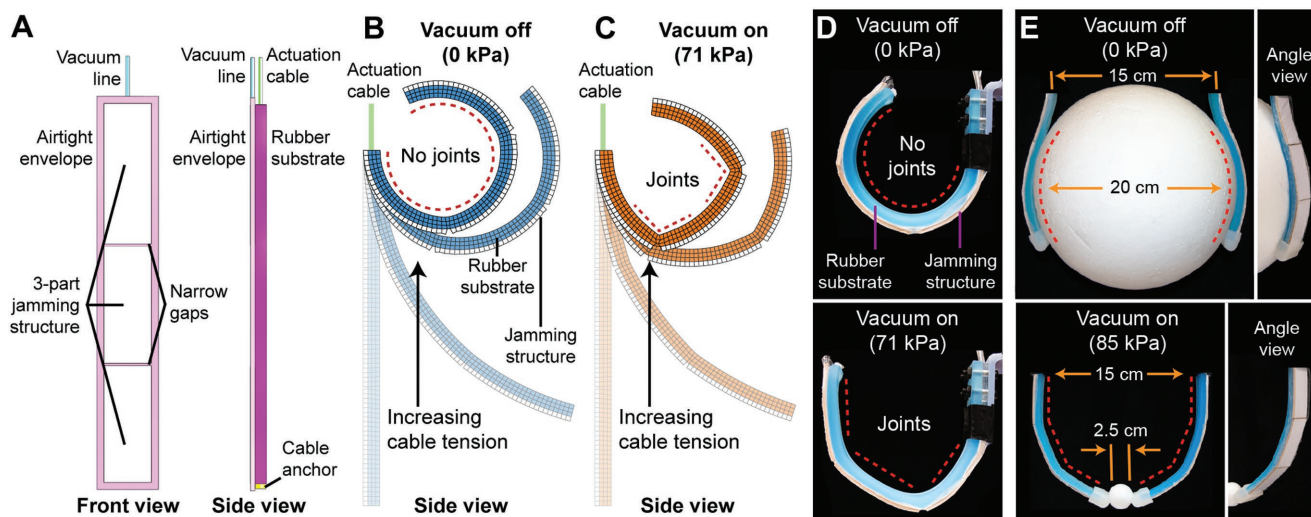


Figure 5. Finite element modeling and experimental demonstration of variable kinematics function. A) Schematic of variable kinematics system. B,C) Finite element simulations of variable kinematics behavior at increasing cable loads. At the highest load, the ratio of maximum to mean curvature increased by a factor of 6.65 with vacuum on, quantitatively verifying the creation of joints. D) Experimental validation of variable kinematics system. E) A two-fingered grasper was constructed in which each finger consisted of a variable kinematics system with a rounded fingertip. When no vacuum was applied to the fingers, the grasper could conform to a large ball, hold it aloft, and resist perturbation, thus performing a stable wrap grasp. When vacuum was applied, the grasper could perform a stable pinch grasp on a ball of one-eighth the diameter.

built a two-fingered grasper in which each finger consisted of a cable-actuated variable kinematics system with a rounded fingertip. When no vacuum was applied and the cables were pulled, the fingers bent continuously, and the device could perform a stable wrap grasp on a ball of diameter 20 cm. When vacuum was applied first, the fingers bent discretely, and the device could perform a stable pinch grasp on a ball of one-eighth the diameter (Figure 5E).

To evaluate the stability of the grasps, multiaxis stiffness measurements were conducted and a perturbation test was performed (SI: Functions and Applications: Two-Fingered Grasper). Stiffness measurements showed that the maximum bending stiffness of a finger increased by at least a factor of 32 when vacuum was applied. Simultaneously, the off-axis bending stiffness (i.e., the stiffness along the perpendicular bending axis) increased by a factor of 2.5, and the torsional stiffness increased by a factor of 2.7. Furthermore, perturbation tests demonstrated that the force required to dislodge the small ball was at least a factor of eight greater when vacuum was applied and the discretely bending fingers were used, compared to when no vacuum was applied and the continuously bending fingers were used. As an additional feature, applying vacuum after the continuously bending fingers performed a grasp increased their resistance to further deformation.

3. Discussion

3.1. Modeling

Earlier studies of laminar jamming exclusively predicted the stiffness of jamming structures during pre-slip, as well as the first transition load (i.e., the load at which the structures move from pre-slip to the transition regime).^[18,21,28] In contrast, our analytical model predicted the elastica, stiffness, energy dissipation, and damping of two-layer jamming structures during pre-slip, the transition regime, and full-slip, as well as determining both the first transition load and the second transition load (i.e., the load at which the structures move from the transition regime to full-slip). Our finite element models of many-layer jamming structures then extended the predictions of the analytical model to structures with arbitrary numbers of layers. Thus, the analytical and finite element models completely described the mechanical behavior of jamming structures over all three phases of deformation.

Together, the models provide designers with an accurate and efficient means to predict the mechanical behavior of arbitrary jamming structures. In particular, no models have existed for mechanical behavior in the transition regime or full-slip. To determine how a particular jamming structure will behave in these phases, designers have had to fabricate and characterize the structure. In our experience, this process requires hours of continuous labor per structure. In contrast, the analytical model can predict experimental behavior for a two-layer jamming structure immediately, and a finite element simulation can predict experimental behavior for a many-layer structure in less than one hour without supervision.

In addition, the functional dependencies of performance metrics on design inputs were extracted for many-layer

jamming structures. These relations provide researchers with a rapid means to meet arbitrary design requirements. For instance, if the full-slip stiffness of a jamming structure must be reduced by a factor of four (e.g., for an orthosis that softens at high loads for user safety), the relations show that the number of layers or vacuum pressure can be reduced by a factor of four, or the coefficient of friction can be reduced by a factor of two (Table S1, Supporting Information). Likewise, if the full-slip damping of a jamming structure must be increased by a factor of four (e.g., for a field robot that dampens impacts to protect components), then the number of layers, vacuum pressure, or coefficient of friction can be increased by a factor of four. The vacuum pressure can be controlled on command with a vacuum regulator; thus, full-slip stiffness and damping can be adjusted in real-time. The finite element models can be used to derive functional dependencies between additional performance metrics and design inputs as desired. Note that although the models and experiments in this study investigated jamming structures with a maximum of 20 layers, the extracted functional dependencies are applicable to structures with arbitrary numbers of layers; however, as the number of layers increases, it may become challenging to physically achieve a homogeneous vacuum pressure throughout the structure.

3.2. Useful Functions

Previous studies applied laminar jamming to diverse applications. However, these studies almost exclusively used laminar jamming to control stiffness; furthermore, when the jamming structures were integrated with actuators, the structures controlled the stiffness of the system while the actuators were continuously powered. We expanded on these capabilities by demonstrating shape-locking and variable kinematics. The former enables soft machines to preserve their shape after the actuation input is removed, whereas the latter enables them to select between continuous and discrete bending.

Shape-locking illustrates one way in which laminar jamming structures can enable soft machines to reversibly emulate traditional rigid robots. Nearly, all traditional robotic arms can navigate to an arbitrary location in their workspace and resist static loading. Furthermore, some arms have brakes that allow them to resist loading after power is disconnected. Shape-locking endows soft machines with precisely this ability, as it enables them to achieve an arbitrary configuration, lock in place, and resist static loading, even after disconnecting the actuation input. Soft machines can thus save power by requiring no control effort to preserve their shape; furthermore, soft machines with high material strain (e.g., McKibben actuators) can be deflated after locking, mitigating the risk of catastrophic rupture. Shape-locking may also be achieved by combining soft actuators with other materials and structures capable of tunable stiffness and damping, such as recently developed thermally activated fibers and composites;^[29–31] however, laminar jamming offers a promising combination of rapid actuation, human safety, and high range of bending stiffness between the deforming and shape-locked states.

Variable kinematics comprises a second way in which laminar jamming structures can link the behavior of soft machines and traditional rigid robots. Specifically, this function can allow soft machines to rapidly transform between a highly compliant state in which they can conform to arbitrary shapes, and a rigid, jointed state in which they can behave like a serial manipulator. As demonstrated in this study, variable kinematics can enhance the performance of robotic graspers. Moreover, this capability can be useful for any device where both conformability and rigidity are desired (e.g., in surgical devices that must traverse vasculature, but subsequently apply high forces).

More generally, variable kinematics facilitates the modeling, sensing, and control of soft machines. For traditional rigid robots, multirigid-body mechanics can describe forward and inverse kinematics; on the other hand, soft machines require the mathematical tools of continuum mechanics, which are generally far more complex. Furthermore, in traditional rigid robots, a small number of sensors can accurately estimate configuration; in soft machines, many sensors are required. Because modeling and sensing is more complex for soft machines, control is inherently more difficult.^[32] Variable kinematics allows soft machines to behave like multirigid-body systems, with rigid links connected by joints. Thus, they can be modeled and sensed like traditional rigid robots, greatly simplifying their control. (It is interesting to note that octopuses use variable kinematics to simplify control, creating joints along their tentacles to facilitate fetching tasks.^[33])

3.3. Limitations

Our modeling and demonstrations have three notable limitations, each of which can be resolved as described. First, in our finite element models of many-layer jamming structures, the execution time of the simulations scaled linearly with the number of layers; for models of jamming structures with exceptionally high numbers of layers, the time may become prohibitive. Nevertheless, as the numbers of layers increases in a jamming structure with a fixed total thickness, the structure may be accurately approximated as a single crystal with a single slip system. This structure can be simulated more simply than a multilayer structure, reducing execution time (SI: Finite Element Modeling: Limiting Behavior).

Second, in our shape-locking demonstration, our prototype still required a vacuum source to be connected after depressurization. Thus, the device would be challenging to operate in environments where supporting equipment is unavailable. This difficulty could be resolved by using a one-way valve to maintain vacuum after the vacuum input is disconnected.

Third, in our demonstrations, vacuum was used to actuate the jamming structures. As a result, the maximum pressure gradient acting on the jamming structures was limited to the absolute ambient pressure, which in turn reduced the maximum load that could be sustained by the structures before their stiffness declined. This limit may be overcome by using electrostatic actuation^[28] or elastic actuation, in which the layers are reversibly compressed by an external elastic structure (e.g., a mesh envelope^[24] or spring clips (SI: Additional Concepts: Spring-Based Jamming)).

4. Conclusion

This paper demonstrates how the nonlinear structural phenomenon of laminar jamming can bridge the paradigms of soft robotics and traditional rigid robotics. We have derived an analytical model for two-layer jamming structures over all major phases of deformation, constructed highly accurate finite element models of many-layer laminar jamming structures, and extracted functional dependencies of major performance metrics on critical design inputs. We have demonstrated two novel functions, shape-locking and variable kinematics, that illustrate how laminar jamming can reversibly endow soft machines with behavior typical of traditional rigid robots. We also built a simple grasper capable of both pinch grasps and wrap grasps, demonstrating how laminar jamming can enhance the performance of real-world soft robotic systems. Collectively, our work elucidates the mechanics of laminar jamming, accelerates the design process of jamming structures, and provides a foundation for creating mechanically versatile machines and structures that cannot simply be categorized as “soft” or “rigid.”

5. Experimental Section

The following is an abridged description of the methods used in this study. For complete detail, see Supporting Information.

Analytical Modeling: The axial strain fields in each layer of the jamming structure were approximated as a superposition of a field that varied linearly with the vertical coordinate and a field that was constant with the vertical coordinate. An interfacial displacement variable was defined. Moment-stress relations and static equilibrium were used to derive governing equations for sections of the structure with cohesive interfaces and sections with slipped interfaces. Boundary conditions were formulated for clamped and free boundaries, and continuity conditions were defined to couple cohesive and slipped interfaces. The boundary-value problem was then explicitly solved to determine the elastica of a cantilevered jamming structure with a uniform distributed load in the pre-slip regime, transition regime, and full-slip regime. During the transition regime, the location of the transition between cohesive and slipped interfaces was also determined. The results were then used to derive stiffness, energy dissipation, and damping in each regime, as well as critical loads between the regimes. Dimensionless parameters were defined to nondimensionalize all results.

Finite Element Modeling: All finite element models were constructed using finite element simulation software (ABAQUS 6.14r2, Dassault Systèmes, Villacoublay, France). In the finite element models of the two-layer and many-layer jamming structures, each layer was approximated as a 2D plane-strain structure. Pressure equal to vacuum pressure was applied to all outer surfaces, and loads were subsequently applied. Large-deformation analysis was enabled, and the interfaces between the layers were defined as contact surfaces with a penalty friction formulation. A uniform mesh was used consisting of square four-node bilinear plane-strain quadrilateral elements with reduced integration. Each layer was meshed with two elements across its thickness.

In the finite element models of the variable kinematics structures, the rubber substrate and each of the jamming structures were modeled as a homogeneous 2D plane-strain structure. To simulate the vacuum-on condition, the elastic modulus of the jamming structure was assigned to that of paper, and to simulate the vacuum-off condition, the modulus was reduced by a factor of n^2 , with $n = 20$ to match experimental conditions. Cable actuation was approximated as a pure moment load. Large-deformation analysis was enabled. A uniform mesh was used consisting of square four-node bilinear plane-strain quadrilateral hybrid elements with reduced integration. The structure was meshed with four elements across its thickness.

Fabrication of Jamming Structures: The jamming structures were fabricated in five distinct steps: (1) Sheets of copy paper (HP Ultra White Multipurpose Copy Paper) were cut into strips on a laser cutter (VLS4.60, Universal Laser Systems, Inc., Scottsdale, AZ). (2) An acrylic frame enclosing the strips was cut on the laser cutter. The height of the acrylic frame was selected to be greater than the total thickness of the strips. The frame was used only for fabrication and did not comprise part of the jamming structure. (3) A sheet of thermoplastic polyurethane (TPU) (American Polyfilm, Inc., Branford, CT) was formed to the acrylic frame on a vacuum former (Formech 300XQ, Formech International Limited, Hertfordshire, UK). (4) The strips of paper and TPU tubing (Eldon James Corp., Denver, CO) were placed into the frame. The TPU sheet was folded over its contents, and the assembly was pressed in a heat press (Powerpress, Fancierstudio, Hayward, CA) at 100 °C. Since the height of the acrylic frame was greater than the total thickness of the strips, the force of the heat press was concentrated on the frame; thus, only the region of the TPU sheet above the frame was sealed. The frame was then removed from the assembly. The remaining components comprised a jamming structure. (5) The end of the structure containing the TPU tubing was sandwiched between two conforming aluminum blocks. The blocks were heated to 171 °C on the heat press, creating a circumferential seal around the tubing. The blocks were then removed from the assembly.

Experimental Characterization: Jamming structures were tested on a three-point bending fixture in a universal materials testing device (Instron 5566, Illinois Tool Works, Norwood, MA). The structures were placed on the fixture and connected to a manual vacuum regulator (EW-07061-30, Cole-Parmer, Vernon Hills, IL) set to the desired pressure. The loading anvil of the testing device was lowered at a rate of 25 mm min⁻¹ until reaching the desired maximum displacement. Force and displacement measurements were simultaneously recorded.

Functions and Applications: All molds were designed using CAD software (Solidworks 2015, Dassault Systèmes, Villacoublay, France) and 3D printed using a stereolithography-based printer (Objet30 Scholar, Stratasys, Ltd., Eden Prairie, MN). For the actuator used in shape-locking demonstrations, a two-part mold was designed, and the actuator was cast from shore 10A platinum-cure silicone rubber (Dragon Skin 10 Medium, Smooth-On, Inc., Macungie, PA). The actuator and jamming structure were bonded using silicone building sealant (Dow Corning 795, Dow Corning, Midland, MI).

For the substrate used in the variable kinematics demonstrations, a one-part mold was designed with an inserted rod to create a channel for an actuation cable. The substrate was cast from high-stiffness PDMS rubber (Sylgard 184, Dow Corning, Midland, MI). The substrate and three-part jamming structure were again bonded using silicone building sealant. The cable consisted of braided polyethylene (Hollow Spectra, BHP Tackle, Harrington Park, NJ) and was tensioned using a turnbuckle mechanism.

For the fingertips of the fingers in the two-fingered grasper, a two-part mold was designed, and the fingertip was cast from shore 00–10A silicone rubber (Ecoflex 00–10, Smooth-On, Inc., Macungie, PA). Multiaxis stiffness tests and perturbation tests were performed using a digital force gauge (Chatillon DF110, AMETEK Sensors, Test & Calibration, Berwyn, PA) and custom-built fixtures.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

The authors would like to thank the members of the Harvard Biorobotics Laboratory; the Vlassak Group; the Harvard Biodesign Laboratory; the Bertoldi Group; and James Weaver, Ph.D., for their technical advice and

assistance. Funding was provided by the National Science Foundation Graduate Research Fellowship Award 1122374 and the National Science Foundation National Robotics Initiative Grant CMMI-1637838.

Conflict of Interest

The authors declare no conflict of interest.

Keywords

laminar jamming, shape locking, soft robotics, variable impedance, variable kinematics

Received: December 7, 2017

Revised: January 18, 2018

Published online:

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